ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS **6032/3**

PAPER 3 THEORY 2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Answer booklet
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.
Write your answers on the separate answer booklet provided.
If you use more than one booklet, fasten the booklets/sheets together.
All working for numerical answers must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

This specimen paper consists of 13 printed pages.

DATA

speed of light in free space	$c = 3.00 \times 10^{8} \mathrm{ms}^{-1}$
permeability of free space	$\mu_o = 4\pi \times 10^{-7}$ Hm^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ Fm^{-1} $(1/4\pi\varepsilon_0 = 8.99 \times 10^{9}$ mF$^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
the Planck constant	$h = 6.63 \times 10^{-34}$ Js
unified atomic mass unit	$1\ u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ JK^{-1}mol^{-1}
the Avogadro constant	$N_A = 6.02 \times 10^{23}$mol^{-1}
the Boltzmann constant	$k = 1.38 \times 10^{-23}$ JK^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$Nm^{2}kg^{-2}
acceleration of free fall	$g = 9.81$ ms^{-2}

FORMULAE

uniformly accelerated motion	$s = ut + ½at^2$
	$v^2 = u^2 + 2as$
work done on/by a gas	$W = p\,\Delta V$
gravitational potential	$\emptyset = -\, Gm/r$
hydrostatic pressure	$p = \rho gh$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_o \cos \omega t$
	$v = \pm\omega\sqrt{(x_o^2 - x^2)}$
Attenuation of x-rays	$I = I_o e^{-\mu x}$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy of charged capacitor	$W = \frac{1}{2}QV$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_o \sin\omega t$
radioactive decay	$x = x_o \exp(-\lambda t)$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$

Answer question 1 and any other 3 from the remaining questions.

(a) **(i)** Express the unit Ohm in terms of base units.

(ii) In a rural electrification project powered by wind energy, the electrical power output, P, of a horizontal axis wind turbine is influenced by the blade length, L, air density, ρ, and windspeed, V.

Using dimensional analysis, derive an equation expressing P in terms of ρ, L and V. [5]

(b) **(i)** State two conditions necessary for equations of motion to be applied.

(ii) During a community cycling event promoting sustainable transport, two cyclists, A and B, compete over a 50 m straight track. They start simultaneously. Cyclist A maintains constant speed at 26 m/s. Cyclist B rides the first 200 m at 15 m/s, then accelerates uniformly at 10 m/s^2 to complete the race.

Determine who wins the race. Show all working.

[5]

(c) **(i)** Define *linear momentum*.

(ii) At a recreational club, two snooker balls P and Q collide on a smooth table. Q of mass 250 g moving with a velocity of 7 m/s collides with P, initially at rest. After collision Q moves with a speed of 2 m/s.

1. Calculate the velocity of ball P after collision.

2. Determine whether the collision is elastic or inelastic.

[5]

(d) **(i)** State the origin of upthrust on a body immersed in a fluid.

(ii) **Fig. 1.1** shows a tourist wearing a life-saving jacket who accidentally fell in a dam.

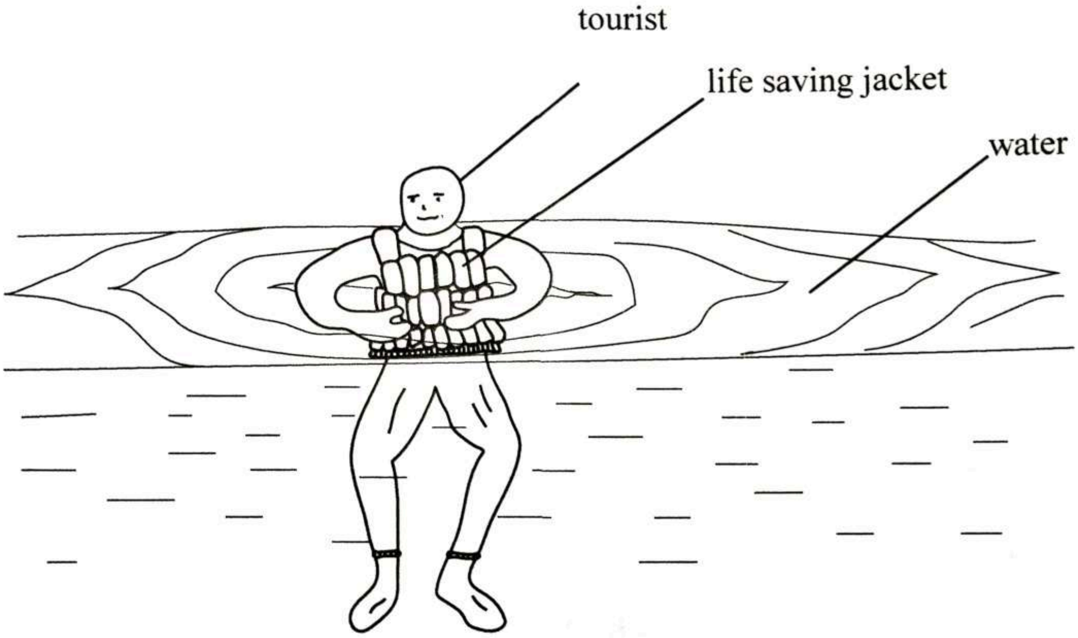

Fig. 1.1

1. Name the forces acting on the tourist.

2. The tourist's life-saving jacket bursts and the tourist starts to sink eventually reaching terminal velocity. Explain, quantitatively how the terminal velocity is reached.

[6]

(e) **(i)** State *Newton's law of gravitation.*

(ii) Derive the equation for the gravitational field strength at a point,

$$g = \frac{GM}{r^2}.$$

(i) Sketch the graph to show the relationship between gravitational field strength, g and distance, r,

1. outside the earth,

2. inside the earth. [4]

2 **(a)** The needle of a sewing machine vibrates with a small displacement, x. **Fig 2.1** shows the variation of displacement, x, against time, t.

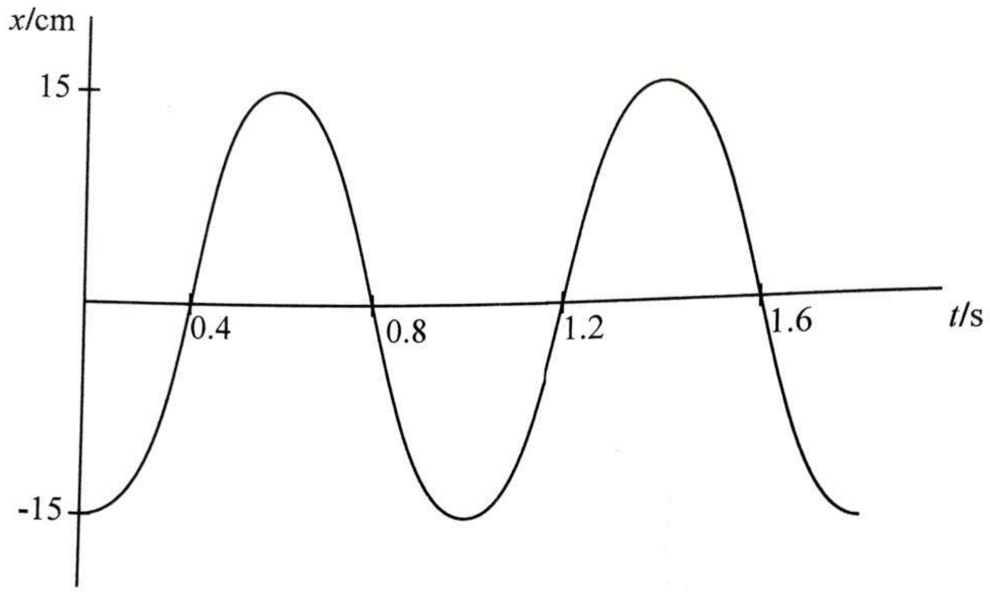

Fig. 2.1

(i) Use the graph to determine the period of oscillation.

(ii) Calculate the

1. frequency,
2. angular frequency,
3. velocity at a displacement of 10 cm,
4. maximum velocity.

(iii) Use the equation of the form $x = x_0 Sinwt$ to determine the displacement at 1.2 μs.

[11]

(b) CT scanning uses multiple X-ray beams and computer processing to create detailed internal body images.

(i) Explain the principles of CT scanning.

(ii) State **two** advantages of CT scan over ordinary X-rays.

[6]

(c) A soccer supporter was blowing a vuvuzela and produced stationary sound waves shown in **Fig. 2.2**.

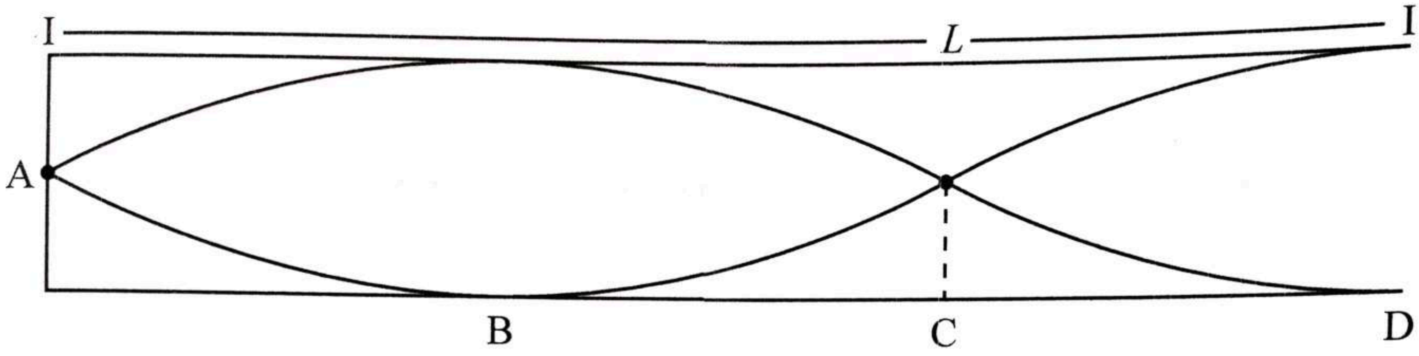

Fig. 2.2

(i) Identify positions of antinodes.

(ii) Calculate the wavelength of the sound waves if the length, L, of the vuvuzela pipe is 45 cm.

[4]

(d) A narrow beam of coherent light is incident normally on a diffraction grating having 4.0×10^5 lines per metre. The number of diffracted maxima on each side is found to be 4.

(i) Explain what is meant by *coherent light*.

(ii) Determine the wavelength of the light wave.

[4]

3 **(a)** **Fig. 3.1** shows a circuit designed by a high school student to switch on a bulb automatically when it gets dark.

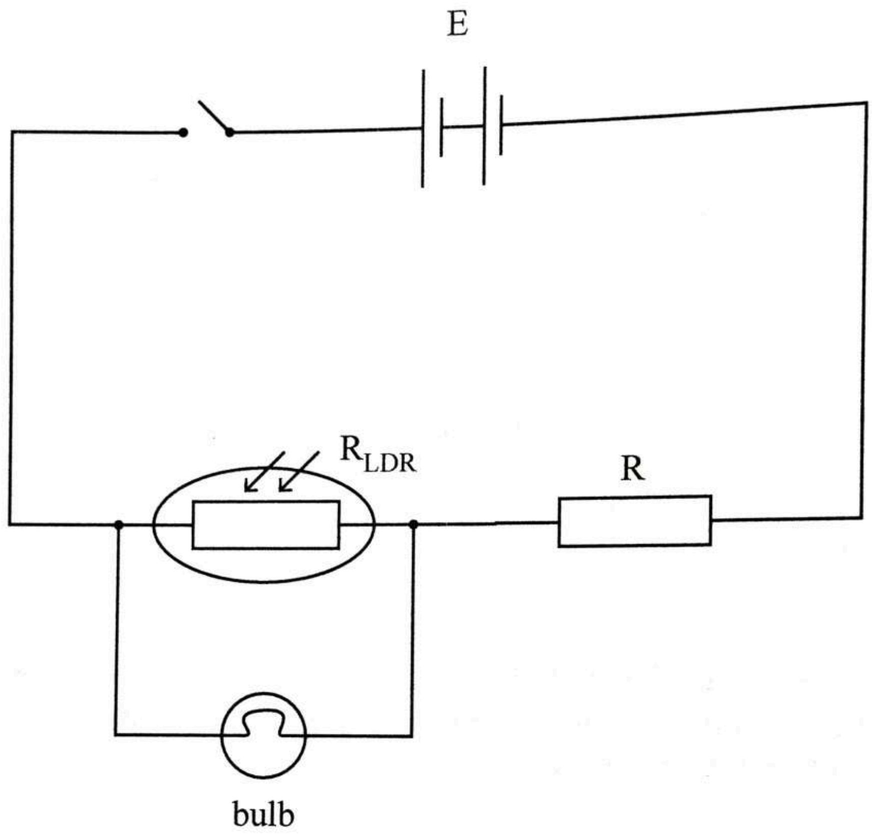

Fig. 3.1

(i) Describe what happens when the switch is turned on.

(ii) Modify the circuit so that the circuit turns on during the day.

[4]

(b) **(i)** Define the *Farad.*

(ii) **Fig. 3.2** shows three 80 μF capacitors connected in series to a 12 kV d.c supply.

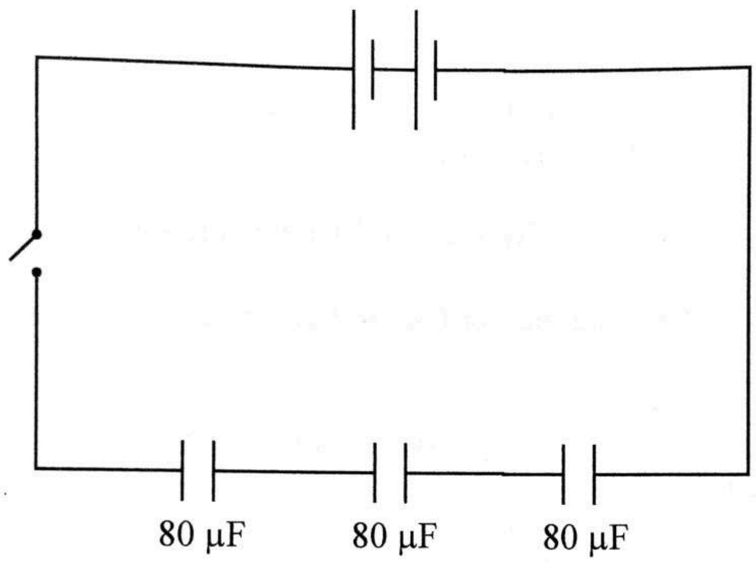

Fig. 3.2

Calculate the total

1. capacitance,

2. charge stored,

5. energy stored.

[7]

(c) **(i)** Define the *Weber*.

(ii) A coil in a magnetic field of flux density 25 mT is 14 cm long and 18 cm wide. Calculate the magnetic flux in the coil.

(iii) Outline an experiment to verify Lenz's law.

[8]

(d) **(i)** A technician develops a wheeled robot to safely deliver pharmaceuticals along a marked path.

1. Name a suitable sensor to detect this path.

2. Explain how the sensor named in **(d) (i)**1. enables the robot to follow the path.

6. The technician used geared d.c motors instead of ordinary d.c motors to drive the wheels.

   State **two** advantages of using geared d.c motors.

7. State a suitable sensor to help the robot to avoid obstacles.

(iii) Write a code to rotate a servo motor named myservo through 90 degrees.

[6]

(a) Define *specific latent heat of vaporisation.* [2]

(b) A farm is irrigated via a 2.4 m wide and 18 km long canal. Water flows at 1.45 m^3/s. During the day, the canal surface absorbs solar power at 700 W m^{-2}.

(i) Explain why more water is supplied than is used by the farm.

(ii) The specific latent heat of vaporization of water is $2.26 \times 10^6 J\,kg^{-1}$ and the density of water is 1 000 kgm^{-3}.

Determine the rate at which water is supplied.

(iii) 1. State any **two** ways to reduce the loss of water from the canal.

2. List the problems that could arise if the solutions in **(iii)** 1 were to be implemented.

[11]

(c) **Table 4.1** show masses of the various molecules in the air that behave ideally and are in thermal equilibrium.

Table 4.1

molecules	carbon dioxide	argon	oxygen	nitrogen
Mass of molecules/u	44	40	32	28

(i) Show that the mean-square speed, $<c^2>$ of the molecules in air is given by $< c^2 > \propto \frac{1}{m}$.

(ii) State which molecule has the smallest root-mean square speed.

(iii) Given that the root mean square speed of nitrogen molecules is 513 m/s.

Determine the root mean square speed of oxygen molecules.

(iv) A dust particle has a mass of 5.3×10^{-14} kg.

1. Calculate the estimated root mean square speed of dust.

2. State an assumption made in **(iv)(1)**.

[8]

(d) State the *fixed points of an absolute scale.* [2]

(e) Derive the equation of continuity Av = constant. [2]

5 **(a)** **(i)** Explain what is meant by *quantisation of charge*.

(ii) A charged oil droplet of density 960 kg m^{-3} is held stationary between horizontal metal plates 3.0 mm apart by a potential difference of 270 V. When the potential is switched off the droplet falls in air with a viscosity of 1.8×10^{-5} Nsm^{-2} and a terminal velocity of 12 mm/s.

Calculate, for the oil droplet, the

1. radius,

2. charge.

(iii) Comment on the value of charge obtained in **(a)(ii)**2. [8]

(b) **(i)** State the *photon energy equation.*

(ii) **Fig. 5.1** shows a vacuum photocell with incident light of wavelength 600 nm directed at the photocell and the ammeter reads 0.15 μA.

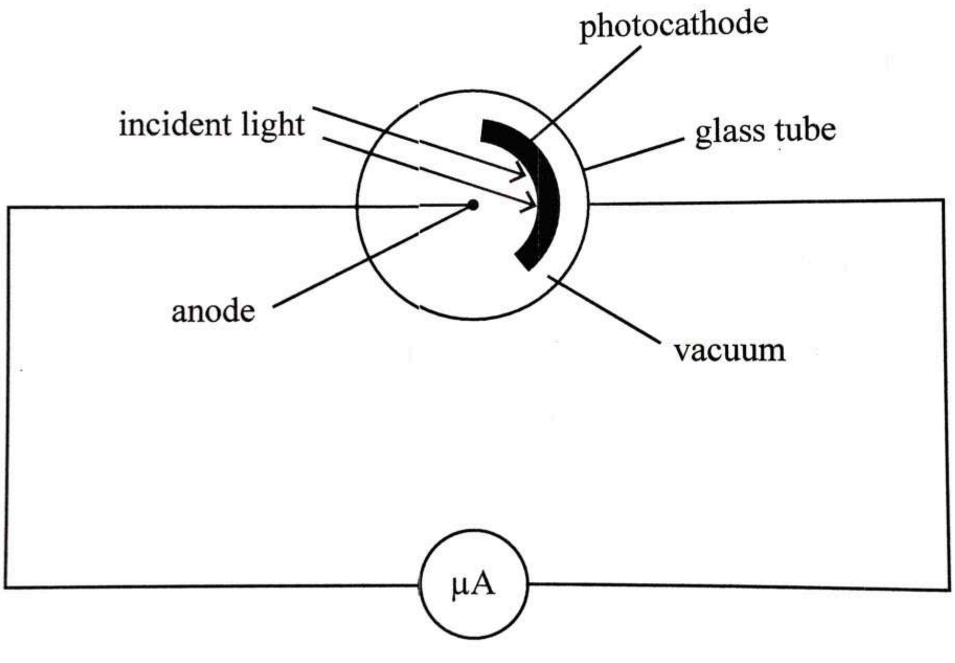

Fig. 5.1

Calculate the

1. energy of incident electromagnetic radiation,

3. number of electrons emitted per second from the photocathode.

(iii) Explain why the ammeter reads zero when light of a longer wavelength is directed at the photocell.

[6]

(c) A natural ore sample from a mine contains strontium-90 which decays spontaneous and randomly with a half-life of 336 months and an activity of 4.6×10^{8} Bq.

(i) Explain *spontaneous* and *random decay*.

(ii) Calculate the decay constant.

(iii) Show that the mass of the sample is 8.6×10^{-8} kg.

[7]

(d) **(i)** Describe the two functions of a copper braid in a coaxial cable.

(ii) State two advantages of transmission of information in digital rather than analogue.

[4]

Blank Page

Blank Page